

AN EVIDENCE AND ESTIMAND AUDIT OF A PAIRED FIXED-FILLER RECORD

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ABSTRACT

A score recorded when a masked-diffusion decoder commits a token and the correctness of its final answer are measurements at different levels. We audit the evidence and estimand behind their cross-level mean difference in two source-reported records. One record gives a natural-strata difference of $+0.2227$; another, on a different reported 400-item pool, gives a paired $+37$ -token, fixed-NFE=64 filler contrast of -0.0075 . These point values concern different samples and estimands, and no joint comparison is defined or tested. Their juxtaposition therefore has no shared inferential interpretation. We independently check only the algebraic estimand definitions and arithmetic printed from the transcribed values. The empirical measurements themselves are source-reported and cannot be independently verified because the current package lacks the item records, exact intervention, decode traces, and analysis implementation. The paired estimand is consequently a descriptive finite-pool quantity conditional on one recorded decode realization, not expected decoder behavior. The commit score is also not verified as a probability of final-answer correctness. No experiment was run as part of this audit.

1 INTRODUCTION

Prompt length can covary with task content, formatting, and difficulty. A natural length association therefore does not identify the effect of a prompt modification. This distinction is especially important for masked-diffusion decoding, where prompt construction, the unmasking trajectory, and a fixed inference budget can interact. We examine one source-described GSM8K (Cobbe et al., 2021) / LLaDA-8B-Instruct (Nie et al., 2025) record that places two analyses side by side: an independent natural-strata analysis and a paired fixed-filler analysis.

The central quantity is a *cross-level mean score–outcome difference*: the mean of a decoder’s recorded token-commit score minus mean final-answer correctness. The source artifacts call this quantity SCE; we retain that string only as an internal artifact alias. We do not call the score confidence or an answer probability because the package contains neither the token-to-answer aggregation nor construct-validation evidence. The quantity remains arithmetically well defined as a directional summary, but its zero point is not a calibration certificate (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024).

This paper is an evidence and estimand audit. It asks what the source-described paired summary estimates, which statements can be checked from the current package, and which empirical conclusions remain unsupported. The natural-strata and fixed-filler point values are catalogued as distinct source records; neither is used to decide what the other means. The paper contributes:

1. a finite-pool estimand conditional on the recorded single-decode realization;
2. a separation between independently checkable definitions and printed arithmetic, on the one hand, and empirical measurements that can only be transcribed, on the other; and
3. a package inventory that makes the unavailable treatment, sample, score, decode, and implementation evidence explicit.

Measurement status. Throughout, SR/CV means *source-reported and not independently verifiable from the current package*. Every empirical numerical value, count, and dose in the manuscript

is SR/CV. This global label also applies when space prevents repetition in individual cells. Evidence IDs A32-E1 and A32-E2 are opaque references to missing experimental records, not claims that those records were recovered or rerun.

2 AUDIT SCOPE AND EVIDENCE BOUNDARY

The audit is deliberately narrower than a study of whether prompt length has a causal effect. We formalize the paired summary conditional on the source-described stored outputs and identify what evidence would be needed to interpret it more broadly. The natural-strata result is a separate contextual record, not a second endpoint or a common-estimand comparator.

Table 1: Audit partition. Every empirical numerical entry is SR/CV.

Audit status	Object	Location	Boundary
Independently checked	Estimand definitions and displayed arithmetic	This manuscript	Internal form only; no empirical input verification
Transcribed only	Natural-strata point record	A32-E1	Observational target on a separate, unavailable sample
Transcribed only	Paired-filler point record	A32-E2	Finite pool and one recorded decode; treatment and rows unavailable

What is missing. The authoring package does not contain the literal filler string, insertion location, tokenizer-level dose check, immutable 400-item manifest or selection rule, overlap map with the natural-strata sample, exact prompt serialization, arm execution order, token-to-answer score construction, or gate implementation. It also lacks per-item records, decode traces, preregistration, analysis code, model/tokenizer revision, environment lock, and a claim-to-file crosswalk. Consequently, the treatment, sampling frame, score construct, and empirical measurements cannot be independently reconstructed. No interval, probability-calibration, causal-effect, or decoder-expectation claim is retained.

3 FINITE-POOL RECORDED-REALIZATION ESTIMAND

Let $\mathcal{P}$ denote the source-described finite paired pool and $a \in \{S, L\}$ the Short and Long arms. Let ω denote the complete recorded single-decode realization, including every stochastic choice used to produce the stored arm outputs. For item i , let $s_{ia\omega}$ be the source-recorded commit score and $y_{ia\omega} \in \{0, 1\}$ final-answer correctness under the source’s matcher. Define the arm summary

$$D_{\mathcal{P},\omega}(a) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} (s_{ia\omega} - y_{ia\omega})$$

and the paired filler contrast

$$\Delta_{\mathcal{P},\omega} = D_{\mathcal{P},\omega}(L) - D_{\mathcal{P},\omega}(S) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} [(s_{iL\omega} - y_{iL\omega}) - (s_{iS\omega} - y_{iS\omega})].$$

This is a descriptive contrast for the named finite pool and stored decode realization. It is not an expectation over a problem population, decoder randomness, filler strings, or prompt constructions. A decoder-expectation estimand would require deterministic replay evidence or multiple paired decode realizations; neither is available.

Short and Long are labels attached to the source summaries. Without the literal prompts, assignment record, pairing rows, and execution trace, this audit cannot verify arm construction, intervention fidelity, or execution order.

The independent natural-strata analysis instead compares high- and low-length groups formed from a different source-reported record. Natural prompt content and difficulty co-vary with those groups. Without item identities or an overlap map, no common-sample analysis, covariance estimate, joint test, or prespecified comparability/equivalence margin is available.

4 SOURCE-REPORTED RESULTS

4.1 INDEPENDENT NATURAL-STRATA CONTEXT

The source reports that the high-minus-low natural-length contrast is $+0.2227$ in the cross-level mean difference. Its component accounting is an identity: the reported score change is -0.0314 and the reported accuracy change is -0.2541 , so $-0.0314 - (-0.2541) = +0.2227$. This verifies transcription arithmetic only, not the underlying measurements.

Table 2: SR/CV independent natural-strata record (A32-E1). The groups are observational and are not the paired filler pool.

Quantity, high minus low	Source-reported point
Mean recorded commit score	-0.0314
Mean answer correctness	-0.2541
Cross-level mean score–outcome difference	$+0.2227$

4.2 PAIRED FIXED-FILLER DIAGNOSTIC

The source describes a 400-item paired comparison in which Long adds a fixed 37-token filler to Short under reported NFE=64. Table 3 transcribes the available summaries. The literal filler, insertion position, tokenizer verification, sample construction, and per-item pairing records are missing, so these labels describe the source record rather than an independently instantiated intervention.

Table 3: SR/CV Short/Long point record (A32-E2). Empirical inputs and unrounded item-level values are unavailable.

Quantity	Short	Long	Long minus Short
Cross-level mean score–outcome difference	-0.0724	-0.0799	-0.0075
Mean recorded commit score	0.630	0.543	-0.0875
Mean answer correctness	0.703	0.623	-0.0800

The source separately reports the paired-filler point -0.0075 and the natural-strata point $+0.2227$. We do not compute a cross-record difference, ratio, or ordering from them. They concern different item sets and estimands, and the package provides neither a common-sample functional nor a joint analysis. Their juxtaposition is an index of two source records, not an empirical comparison.

The source-displayed score arm means have three decimal places, whereas the separately reported score contrast has four. Without unrounded records, subtraction of the printed means need not recover that contrast. This is unresolved rounding opacity, not evidence of a numerical inconsistency. The current package supports only an audit of the formal target and evidence boundary; it cannot establish intervention fidelity or explain the natural length association.

5 CLOSEST NEIGHBORING WORK AND BOUNDARY

Prior work establishes that prompt choices can change calibration-related behavior, but the verified neighboring studies do not test this exact operation. Xia et al. (Xia et al., 2025) study calibration across four prompt styles; their analysis does not isolate prompt length, content-free filler, masked-diffusion commit scores, or a same-item +37-token intervention. Zhao et al. (Zhao et al., 2024) study six prompting strategies and propose a confidence-calibration prompting method; the prompting strategy and elicited confidence change, rather than the source-described fixed filler under fixed diffusion compute. Kamoda et al. (Kamoda et al., 2023) show that prompt variations affect factual-probing outputs and evaluate prompt augmentation as a calibration intervention; they use prompt variants and ensembling for factual probing, not the present fixed-NFE masked-diffusion filler estimand.

For measurement context, signed mean score–label summaries and their cancellation limits predate this study (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024). The present contribution boundary is therefore an evidence and estimand audit of one source-described record, not a new metric, a general theory of prompt length, or a verified answer-calibration result. A broader prompt-length-specific and masked-diffusion uncertainty search remains incomplete.

6 LIMITATIONS, AVAILABILITY, AND DISCLOSURE

Scientific limits. The record covers one reported model–task setting, one fixed filler description, one dose, one inference budget, and one stored decode per item–arm. The exact sample and intervention are missing. The observational and paired analyses use different targets; the score construct is not verified as an answer probability; and decoder-run variation is unavailable. These limits preclude causal, decoder-average, cross-model, cross-task, or transport conclusions.

What the current package contains. The mutable authoring directory contains `main.tex`, `refs.bib`, the ICLR style files, and compiled manuscript and auxiliary files. It contains no figure asset, and this manuscript includes no figure. The directory does not contain experimental records A32-E1 or A32-E2, literal prompts, item manifests, preregistration, protocol or analysis code, model/tokenizer revisions, decode configuration, an environment lock, or an evidence-ID crosswalk. No empirical value was rerun or independently recomputed.

AI assistance disclosure. AI tools assisted with language editing, $\text{\LaTeX}$ restructuring, and isolated internal pre-submission checks. The authors remain responsible for every claim and evidence boundary. AI assistance did not supply experimental records, generate new measurements, verify the source-reported values, or create scientific figures.

7 CONCLUSION

This audit formalizes a descriptive finite-pool, recorded-single-decode estimand for the source-reported filler point -0.0075 . A separate natural-strata record reports $+0.2227$. The samples and estimands differ, and no joint analysis is available, so the two point values support no cross-record inference. The estimand definitions and printed decomposition arithmetic can be checked independently; the empirical inputs can only be transcribed. Missing records, intervention and sample construction, score semantics, and decoder-replay evidence prevent broader interpretation.

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